

练习七 流体力学（一）

1. h 、 P 、 v

2. P 、 v

3. (3)

4. (4) 要注意已知的是横截面的半径不是面积

5. (1) 粗细两处的流速分别为 v_1 与 v_2 ；则 $Q = S_1 v_1 = S_2 v_2$

$$v_1 = \frac{Q}{S_1} = \frac{3000 \text{ cm}^3 \cdot \text{s}^{-1}}{40 \text{ cm}^2} = 75 \text{ cm} \cdot \text{s}^{-1} = 0.75 \text{ m} \cdot \text{s}^{-1} \quad ;$$

$$v_2 = \frac{Q}{S_2} = \frac{3000 \text{ cm}^3 \cdot \text{s}^{-1}}{10 \text{ cm}^2} = 300 \text{ cm} \cdot \text{s}^{-1} = 3 \text{ m} \cdot \text{s}^{-1}$$

(2) 粗细两处的压强分别为 P_1 与 P_2

$$P_1 + \frac{1}{2} \rho v_1^2 = P_2 + \frac{1}{2} \rho v_2^2$$

$$\Delta P = P_1 - P_2 = \frac{1}{2} \rho v_2^2 - \frac{1}{2} \rho v_1^2 = \frac{1}{2} \times 10^3 \times (3^2 - 0.75^2) = 4.22 \times 10^3 \text{ (Pa)}$$

(3) $(\rho_{\text{水银}} - \rho_{\text{水}}) \cdot g \times \Delta h = \Delta P$ ； $\Delta h = 0.034 \text{ m}$

也可近似 $\rho_{\text{水银}} g \times \Delta h = \Delta P$ ； $\Delta h = 0.032 \text{ m}$

6. (1) 射程 $s = vt$

$$\because \frac{1}{2} \rho v^2 = \rho gh \quad \therefore v = \sqrt{2gh} \quad \text{又} \quad H - h = \frac{1}{2} g t^2 \quad t = \sqrt{\frac{2(H-h)}{g}}$$

$$\therefore s = vt = \sqrt{2gh} \cdot \sqrt{\frac{2(H-h)}{g}} = 2\sqrt{h(H-h)}$$

(2) 设在离槽底面为 x 处开一小孔，则同样有：

$$\frac{1}{2} \rho v_1^2 = \rho g(H-x) \quad v_1 = \sqrt{2g(H-x)} \quad \text{又} \quad x = \frac{1}{2} g t_1^2 \quad t_1 = \sqrt{\frac{2x}{g}}$$

$$\therefore s_1 = v_1 t_1 = 2\sqrt{x(H-x)} = s = \sqrt{h(H-h)} \quad \therefore x = h$$

则在离槽底为 h 的地方开一小孔，射程与前面相同。